

Varun Tulsian

AI Engineering Leader | Leading engineering at Fermi.ai

0→1 GenAI, Retrieval, Agent Systems, Evals

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Summary

I build AI systems that help customers produce outcomes, not just answers. My pattern is eval-first: scope small, ship a thin slice, measure, then scale, balancing reliability and cost-efficiency. At Fermi.ai I lead the engineering team building an AI tutor for school students, where much of the work is after release: judging live sessions, catching failures and routing them back to fixes. At Needl.ai I grew a lean org and shipped agentic workflows that turned long-form finance docs into analyst-ready artifacts with citations and trust scores. Previously at Walmart, I tied ranking/personalization to retail KPIs across marketplace and search. Hold a high bar on user-facing reliability (no silent errors, verifiable citations); move fast on non-critical internals.

Selected Outcomes

- **AskNeedl (Enterprise QA, on-prem):** ~85% precision, 55–31 vs OpenAI on complex finance QA (internal eval); joint #1 on Kensho; enabled analysts to draft credit notes and extract covenants with citations from up to 2.5k-page PDFs.
- **Custom Reports (credit rating, capital markets):** Idea → POC → deploy in ~3 months; 80%+ coverage of 120+ criteria (rest partial), citations, trust score; analyst-ready UX and ROI case.
- **Org/Systems:** Grew AI team 2→7; instituted eval harnesses, cost-efficiency guardrails, decision records, and clear ownership to avoid diffused responsibility.

Work Experience

AI Engineer, leading the engineering team

Fermi.ai | Bengaluru • Oct 2025 – Present

Scope: AI tutor for school students.

- **Post-release loop:** judging live tutor sessions, catching failures and routing them back to fixes; built agents that run that loop and monitor production traces.
- **Decision models:** models like Jev help reduce the tutor's cost and improve throughput.
- **Public demo:** Fermi's tutor teaching the PageRank part of my PageRank → word2vec → attention post (video on my blog, Oct 2026).

VP, Head of AI

Needl.ai | Bengaluru • Apr 2023 – Aug 2025 (Head of AI from Apr 2023; VP from Apr 2025)

Scope: Led strategy and execution for new AI systems, focusing on evaluation, reliability, and tangible product outcomes for clients.

- **Custom Reports (credit rating, capital-markets reports):** Single-threaded owner from idea → POC → deploy; designed typed outputs, citations, and a criterion-level trust score; partnered with design for analyst-ready UX; delivered the business case that unlocked deployment.
- **AskNeedl (long-doc QA):** Shifted focus from “answers” to outcomes/analyst artifacts; established internal evals, typed outputs to prevent silent errors, and source attribution for trust; achieved benchmark-leading performance.
- **Org & operating system:** Grew AI team 2→7; instituted clear ownership and eval-first scoreboards; individualized coaching (direct, fast feedback) in a light environment with high accountability.
- **Technical depth:** Retrieval and long-doc QA; fine-tuning small LMs, bi-encoders, cross-encoders; parallel LLM inference (TorchServe/vLLM); layout-aware PDF parsing; structured generation; agentic orchestration; on-prem deployment.

Founder, stockOMJ (investing tool) • AI Advisor, IndiLeap

Sep 2025 – Dec 2025

Professional Development — Career Break

Oct 2022 – Mar 2023

- Deepened expertise in deep learning and generative AI; built diffusion-model tutorials on 2D datasets and EMNIST for graduate-level learning.
- Work: [GenAI Image Model Tutorials](#) • Repo: [GitHub](#) (60+ stars; used by grad students)

Senior Manager, Data Science (Search Marketplace)

Walmart Global Tech | Bengaluru • Apr 2020 – Sep 2022

- **Assortment gap analysis** → **\$ impact**: Built semantic search/embedding pipeline; enabled \$20M incremental assortment sales.
- **Search conversion** → **KPI lift**: Led multi-objective ranking; increased search conversion by +61 bps, orders by +1.9%.
- **Seller recommenders** → **adoption**: Launched NLP/transformer recommenders adopted by 7K sellers across 400K products.

Staff Data Scientist (Lead, Personalization)

Sensara Technologies | Bengaluru • Jun 2019 – Apr 2020

- Served ~1M events/hour for 300K+ users; embeddings + knowledge graph models powering the “+You Page.”

Staff Data Scientist (Search)

Walmart Global Tech | Bengaluru • Aug 2013 – May 2019

- **Grocery relevance**: Applied CNN intent/attribute models to lift conversion.
- **Sam's Club personalization**: Developed Bayesian reorder models, increasing conversion by ~+1%.
- **Clickstream ranking**: Led offline/online calibration efforts, resulting in a +1.9% increase in orders.

Research Intern (Software Verification)

Microsoft Research | Bengaluru • Jan 2013 – Mar 2013

- Co-authored MUX, a foundational paper on algorithm selection for model checkers, with the Software Verification team.

Education & Achievements

ME, Computer Science — Indian Institute of Science (IISc) | Bengaluru • Aug 2011 – July 2013

- Top 10% of cohort; CGPA 6.8/8
- Secured AIR 8 (100th percentile) in GATE Computer Science among 136,000 candidates

Selected Publications, Patents & Writing

- Patent-pending: Reliability for long-doc QA (enhanced context retrieval for RAG)
- MUX (algorithm selection for model checkers)
- MintHint (automated repair hints)
- [From PageRank to word2vec to Attention](#) (Oct 2026), with an AI tutor session video
- More available at: <https://varun-ml.github.io>